Furkan Utku Nalbant

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High school student with a TÜBİTAK 2204-A research project in polymer biomaterials. Interested in materials science, biomedical, mechanical and industrial engineering and in sustainable technologies. Licensed basketball player and wildlife photographer.

EDUCATION

Sivas Cumhuriyet University Foundation Science High School
2025 – present

Sivas, Türkiye

- GPA above 96/100; focus on mathematics and natural sciences
- Merit-based school scholarship and athlete scholarship

RESEARCH

TÜBİTAK 2204-A High School Students Research Projects
Submitted 2026

Production of Hernia Meshes: Polyurethane Acrylate by UV Polymerisation and Antimicrobial Coating with Benzalkonium Chloride · team member

- Developed porous hernia-mesh prototypes from biocompatible polyurethane acrylate (PUA) by UV polymerisation and templating; pore size about 250 µm
- Applied a benzalkonium chloride antimicrobial coating by dip coating to reduce surgical infection risk
- Tested antimicrobial performance by culturing the meshes on Tryptic Soy Agar (TSA) and Sabouraud Dextrose Agar (SDA); no microbial growth was observed on the coated meshes
- The UV-curing and templating process proved straightforward and easy to reproduce

UV-curing experimental project
2025 – present

School project group

- Hands-on laboratory work on UV curing and materials testing

ACTIVITIES

Licensed basketball player, shooting guard
Age 8 – present

Cumhuriyet University Foundation College Basketball Club (CVK) youth teams · Sivas league · Kaan Karasu Basketball School

- Won championships and medals in the U11, U12, U14 and U16 categories: three first-place and two second-place finishes

Wildlife and bird photographer
2022 – present

Canon 60D, Canon EF 400mm f/5.6 · Instagram @furkan.utku_wildlife

- Photographs exhibited at the Animal Protection Day photography exhibition, 5 October 2026

3D design and 3D printing

- Measure objects with a micrometre caliper, model them as STL files in beginner 3D software and print them on a 3D printer

Theatre and drama
2025

- Member of school theatre groups and drama projects

SKILLS AND LANGUAGES

Languages	Turkish (native); English (B1)
Laboratory	UV polymerisation, antimicrobial coating, biomaterials research, sample preparation, microbiological culture
Technical	3D modelling (STL), 3D printing, precision measurement
Personal	teamwork, problem-solving, scientific enquiry, time management

ACADEMIC INTERESTS

Materials science, biomedical engineering, polymer engineering, mechanical engineering, industrial engineering, sustainable materials and medical technologies